Velocity-Time Graphs

Exam code: 9MA0

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Velocity-time graphs

What is a velocity-time graph?

- **Velocity- time** graphs show the **velocity** of an object as it moves in a **straight line**
- They show **velocity** (on the vertical axis) against **time** (on the horizontal axis)
- Velocity-time graphs can go **below the horizontal axis** whereas **speed-time graphs** can not

What are the key features of a velocity-time graph?

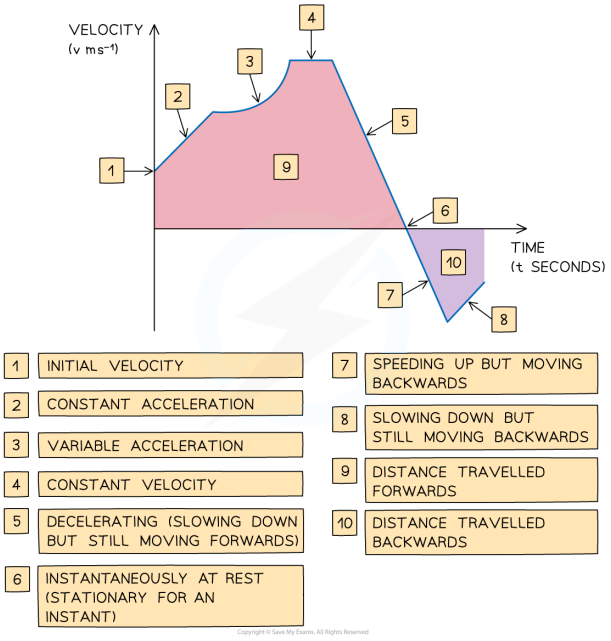
- The **gradient** of the graph equals the **acceleration** of an object
- A **straight line** shows that the object is **accelerating** at a **constant rate**
- A **horizontal** line shows that the object is moving at a **constant velocity**
- The **area** between the graph and the x-axis tells us the **change in displacement** of the object
  - Graph **above** the x-axis means the object is moving **forwards**
  - Graph **below** the x-axis means the object is moving **backwards**
- The **total displacement** of the object from its starting point is the sum of the **areas above** the x-axis **minus** the sum of the **areas below** the x-axis

Help me understand

Stuck on something?
Ask questions and get help with exactly what you're revising.
Got it

stationary at that time

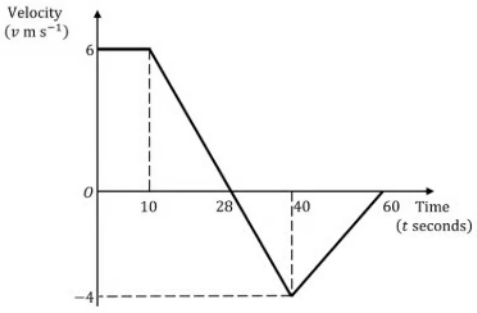
- If the graph is **above** the **x-axis** then the object has positive velocity and is **travelling forwards**
- If the graph is **below** the **x-axis** then the object has negative velocity and is **travelling backwards**



Worked Example

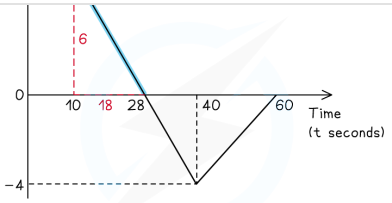
The motion of a bird travelling in a straight line is tracked for 60 seconds.

This information is displayed in the velocity-time graph below.



(a) Calculate the acceleration of the bird at 20 seconds.

Answer:



Acceleration = gradient

Acceleration = $-\frac{6}{18}$ negative as gradient is negative

Acceleration = $-\frac{1}{3}$ ms^{-2} = −0.333 ms^{-2}

Remember units!

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(b) Calculate the distance travelled in the first 28 seconds.

Answer:

b) Displacement = area

Step 1: Identify the shape

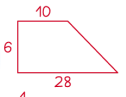
Step 2: Identify the lengths

Step 3: Find the area above x−Axis

So distance = displacement

Distance = 114m

Trapezium



$\frac{1}{2}(10 + 28)(6) = 114$

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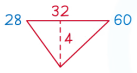
(c) Calculate the displacement of the bird from its starting point after 60 seconds.

Answer:

c) Distance forwards = 114 m

Step 1: Find the distance backwards by finding the area beneath the x−axis



Area = $\frac{1}{2}(32)(4) = 64$ m

THE BIRD HAS TRAVELLED BACKWARDS 64 m

Step 2: Calculate displacement by distance forwards − distance backwards

114 − 64

Displacement = 50 m from starting point

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Examiner Tips and Tricks

- Be careful to spot if you are working with a distance-time graph or a velocity-time graph.
- Be careful to spot if you are working with a speed-time graph or a velocity-time graph.
- Check where the graph starts from on the y-axis, the velocity does not have to start at 0. For example, the scenario could be a car driving at a constant speed and the driver sees a hazard.
- Be extra vigilant when working with negative gradients or with graphs under the x-axis, it is easy to make mistakes with these.
- Speed is a scalar so it can **not** be negative whereas velocity can be.
  Make sure your units are consistent.

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